

DATA 498D

Reflection Essay III

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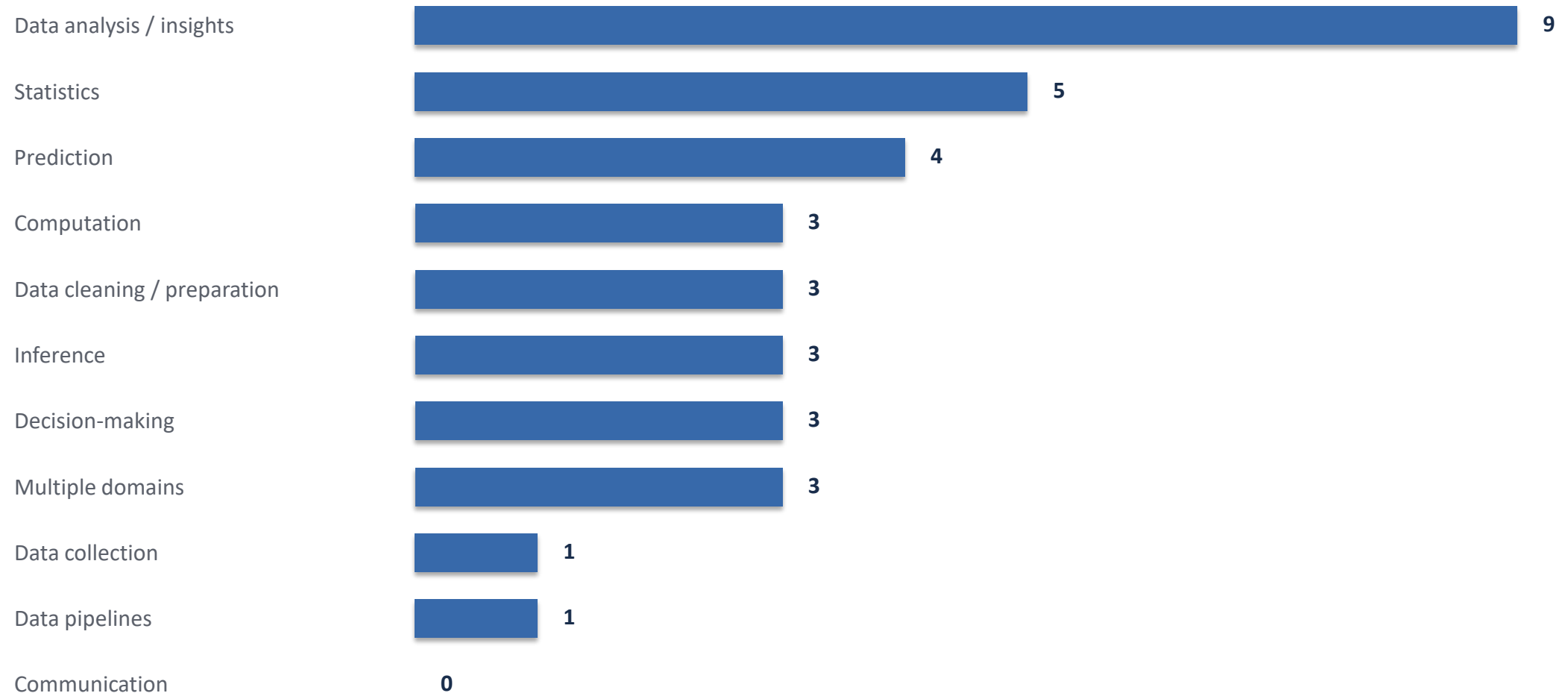


What Is Data Science?

A summary of 10 student definitions

Senior Capstone • Statistics & Data Science

What did students emphasize?



What the class understands well

1. Data → Insights

Most students see Data Science as analyzing data to obtain useful information.

2. Statistics

Many students recognize inference, hypothesis testing, and statistical methods.

3. Prediction

Several students identify prediction as an important outcome of data analysis.

4. Data Science Life Cycle

Some students recognize collecting, cleaning, processing, and managing data.

Where are the gaps?



Data Science $\neq$ Statistics

Some definitions are too centered on hypothesis testing or statistical methods.



Prediction is not the only goal

Data Science also supports explanation, inference, discovery, and decisions.



Computation matters

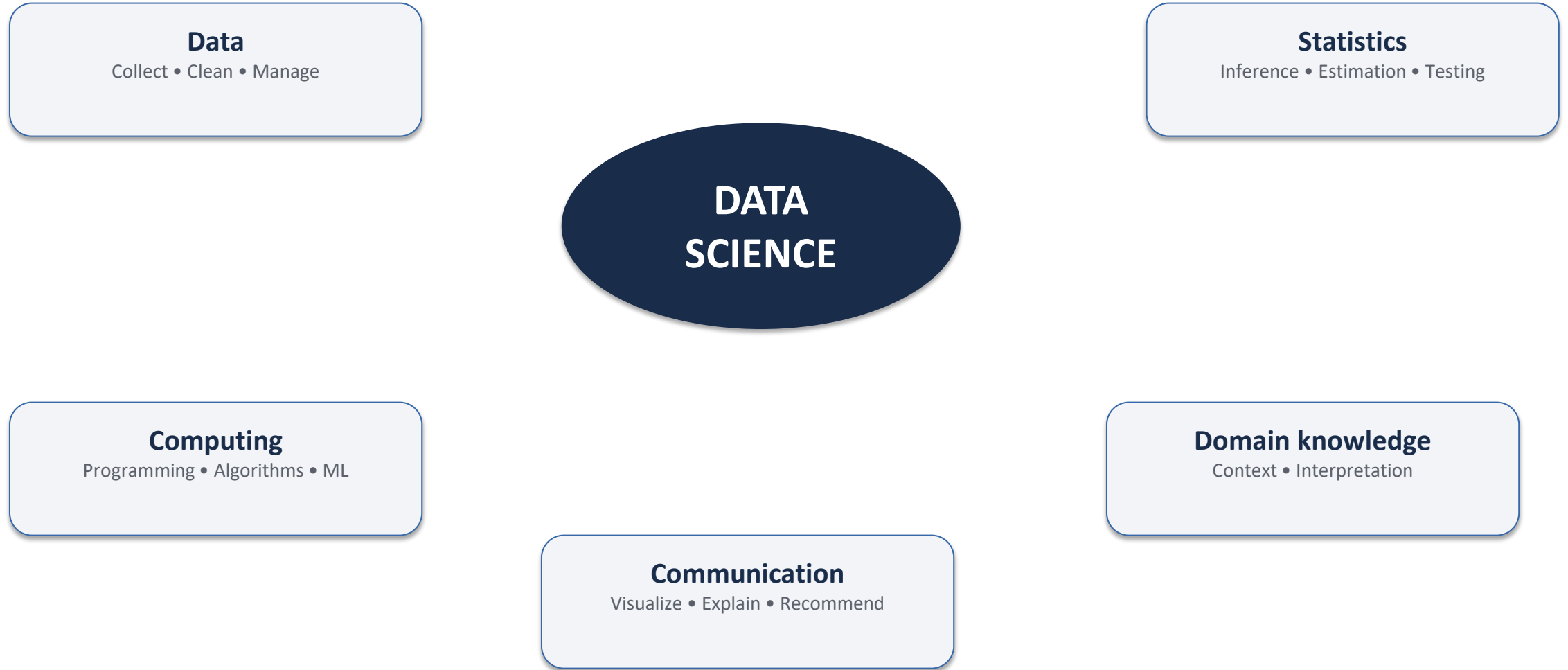
Programming, algorithms, data processing, and scalable workflows are important.



Communication is essential

Students rarely mention visualization, written communication, or explaining results to others.

A broader view of Data Science



Key message for the class

Data Science is broader than statistics.

Data Science is broader than machine learning.

Data Science is broader than programming.

A strong Data Science project connects data, statistics, computing, and domain knowledge.

The final step is communication: explain what the results mean and why they matter.

Appendix

Question: What is Data Science?

Answers from 10 Students:

1. (a) Capabilities: Use and develop statistical models via creating hypothesis testing and present results of the tests. (b) Field application: all applicable, general term, you just have to use hypothesis and scientific method on field data.
2. Data Science is the study of using computational methods to analyze data. Data science also includes making predictions based on data analysis.
3. The use of information to construct meaningful inference or prediction using statistical methods
4. Uses analytical & statistical methods to produce answers to various questions in several different fields, from raw data.
5. A data scientist is someone who has the tools and knowledge to manipulate and work with data to extract information from it: we clean data, apply techniques & once we get our results. We make an educated inference our prediction, can be used in different domains.
6. The study of data, from how the data is collected, to how it is used to make informed decisions
7. A field that utilizes data to derive ideas/information to further another field. It requires a variety of data analysis tools/ideas.
8. A field of study that utilizes raw data to process, translate, and analyze it to get information that raw data contains, so we can predict the future.
9. A study of data in various fields, including a variety of things such as data cleaning, mining, analysis, building pipelines, etc. It helps get insight from raw data.
10. Using computational analysis to discover or make decisions based on data and analyzing how things work.

Reflection question

After seeing your classmates' definitions:

**“What would you add, remove,
or change in your original definition of Data Science?”**

Recommended definition

“Data Science is an interdisciplinary field that combines statistical methods, computational techniques, and domain knowledge to collect, manage, analyze, and model data in order to extract meaningful insights, make predictions or inferences, and support informed decisions.”